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FACULTY OF ENGINEERING

EE 3016 Fundamentals of Electromagnetics

Instructor: Dr. Öğr. Üyesi Mohammad Alsunaidi

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LAB PROJECT 1

Name Surname	Student ID	Department	Sign
Taner Kahyaoğlu	150721015	Electrical-Electronics Engineering	
Mehmet Ali Kamalak	150720062	Electrical-Electronics Engineering	
Muhammet Emre Memili	150721033	Electrical-Electronics Engineering	
Begüm Satıcıoğlu	150723825	Electrical-Electronics Engineering	

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1. NUMERICAL SOLUTION OF POISSON'S EQUATION (REFER TO SLIDES 12-14 WEEK 3)

The main purpose of this simulation project is to numerically solve a boundary-value problem involving the electric potential. The electric potential distribution inside a given structures is obtained numerically. The finite-difference (FD) method is used to solve Poisson's equation in two dimensions.

2. INTRODUCTION

The electric potential distribution inside any given structure can be analytically obtained by solving Poisson's equation subject to some boundary conditions. If we assume no volume charge inside the structure, Poisson's equation becomes (Laplace's equation)

$$\Delta^2 V = 0 \tag{1}$$

In two dimensions (rectangular coordinates), equation (1) is written as

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial x^2} = 0 \tag{2}$$

The ability to solve equation (2) depends in a great deal on the nature of the structure under consideration. In many cases, a closed-form (analytical) solution to equation (2) is difficult to obtain. Alternatively, numerical methods can be used, especially for large problems. Numerical methods utilize the speed of computers and the flexibility of programming.

In this project, the Finite-Difference (FD) method will be used. The FD method is one of the most popular numerical methods in the field of electromagnetics. Its main advantages are (1) easy to formulate, (2) suitable for many structures and (3) flexible for modifications.

The FD-based solution is performed in the following steps.

1) Discretize the given structure (make a grid) using a suitable step size in both dimensions (Δx and Δy). The accuracy of the results improves with smaller step sizes.

2) Approximate the partial derivatives in equation (2) by the following second-order relations.

$$\frac{\partial^2 V}{\partial x^2} = \frac{V(i+1,j) - 2V(i,j) + V(i-1,j)}{\Delta x^2} \tag{3}$$

$$\frac{\partial^2 V}{\partial y^2} = \frac{V(i,j+1)-2V(i,j)+V(i,j-1)}{\Delta y^2} \quad (4)$$

where i and j are the indices along the x- and the y-axis, respectively. Substituting equations (3) and (4) in equation (2), and using $\Delta x = \Delta y$, we get

$$V(i,j) = \frac{1}{4}[V(i+1,j) + V(i-1,j) + V(i,j+1) + V(i,j-1)] \quad (5)$$

3) Solve the resulting linear system of equations for the unknown potential nodes.

Example:

Find the potential distribution inside the structure given in figure 1. Take a step size of 5 cm in both dimensions. The boundary conditions are as shown.

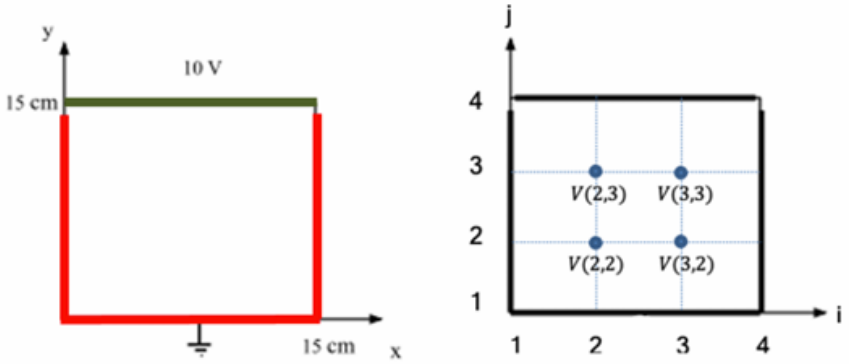


Figure 1 The structure of the example and the corresponding grid.

The grid is arranged as shown in figure 1. At each node in the figure, the value of the potential is labeled. Only four unknown values are to be determined; the potential value for all other nodes are given as boundary conditions (for example, $V(1,1)=0$, $V(4,3)=0$, $V(4,4)=10$).

Applying the algorithm of equation (5), we get the following system of linear equations.

$$4V(2,2) = V(3,2) + V(1,2) + V(2,3) + V(2,1)$$

$$4V(2,3) = V(3,3) + V(1,3) + V(2,4) + V(2,2)$$

$$4V(3,2) = V(4,2) + V(2,2) + V(3,3) + V(3,1)$$

$$4V(3,3) = V(4,3) + V(2,3) + V(3,4) + V(3,2)$$

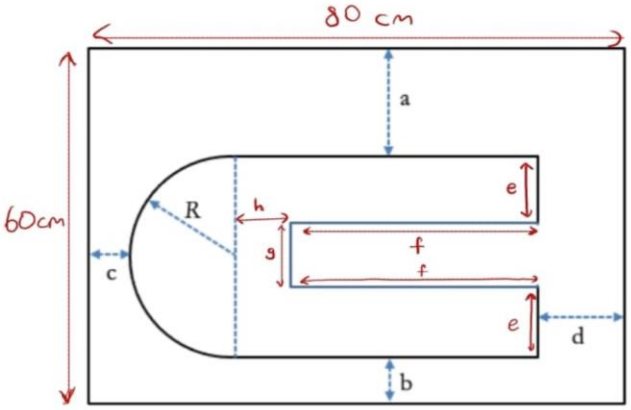
The points in red are boundary (known) points. The system of equations above can be solved (for example) by iteration. This operation can be easily implemented in MATLAB. The result is:

$$V(2,2) = 1.25\,V, \quad V(2,3) = 3.75\,V, \quad V(3,2) = 1.25\,V, \quad V(3,3) = 3.75\,V$$

3. REQUIRED WORK

3.1 QUESTION 1

Solve Laplace's equation using the FD method for the structure given below (figure 2). The fixed structure parameters are given in the table in figure 2. The outer boundary of the structure is biased at +10 V, whereas the inner boundary is biased at -1 V. Use a grid step size of 2 cm in both directions.



Parameter	Value (cm)
a	30
b	10
c	10
d	20
R	10

e = 8
f = 30
g = 4
h = 10

Figure 2 Structure under study.

3.2 QUESTION 2

Produce a contour plot for the equi-potential lines inside the structure. (In MATLAB, there is a function for contouring.)

3.3 QUESTION 3

Produce a vector plot of the electric field inside the structure.

3.4 MATLAB CODING OF QUESTION 1,2,3:

```
clc; clear; close all;


dx = 2; % Grid step in cm
dy = 2;
x = 0:dx:80;
y = 0:dy:60;
[X, Y] = meshgrid(x, y);

% potential matrix
V = 10 * ones(length(y), length(x)); % Entire area with 10V
initially

% boundary conditions for the closed region (-1V)
theta = linspace(pi/2, 3*pi/2, 100); % Semi-circle (counter-
clockwise direction)
r = 10; % Radius
```

```
x_circle = 20 + r * cos(theta);
y_circle = 20 + r * sin(theta);

x_closed = [x_circle, 20, 60, 60, 30, 30, 60, 60, 20];
y_closed = [y_circle, 10, 10, 18, 18, 22, 22, 30, 30];

% Identify the points inside the closed region
mask = inpolygon(X, Y, x_closed, y_closed);

% Apply boundary conditions
V(mask) = -1; % Inside the closed region, potential is -1V
V(~mask & (X == 0 | X == 80 | Y == 0 | Y == 60)) = 10; % Boundary
conditions on outer edges

% Laplace's equation using iterative FD method
tolerance = 1e-4;
max_iter = 5000;
error = Inf;
iter = 0;

while error > tolerance && iter < max_iter
    V_old = V;

    % Update potential using finite difference (excluding boundary
conditions)
    for i = 2:length(y)-1
        for j = 2:length(x)-1
            if ~mask(i, j) % Update only for points outside the
closed region
                V(i, j) = 0.25 * (V_old(i+1, j) + V_old(i-1, j) +
V_old(i, j+1) + V_old(i, j-1));
            end
        end
    end

    error = max(max(abs(V - V_old)));
    iter = iter + 1;
end

% electric field
[Ex, Ey] = gradient(-V, dx, dy);

% Plot the potential distribution with contour lines
figure;
contourf(X, Y, V, 20, 'LineColor', 'none');
colorbar;
title('Equi-potential Lines');
xlabel('x (cm)');
ylabel('y (cm)');
axis equal;
hold on;

% Overlay the structure boundary
plot(x_closed, y_closed, 'k-', 'LineWidth', 1.5);

% Plot the electric field as a vector plot
figure;
quiver(X, Y, Ex, Ey, 'b');
title('Electric Field Countor Plot');
```

```
xlabel('x (cm)');
ylabel('y (cm)');
axis equal;
hold on;

% Overlay the structure boundary
plot(x_closed, y_closed, 'k-', 'LineWidth', 1.5);
```

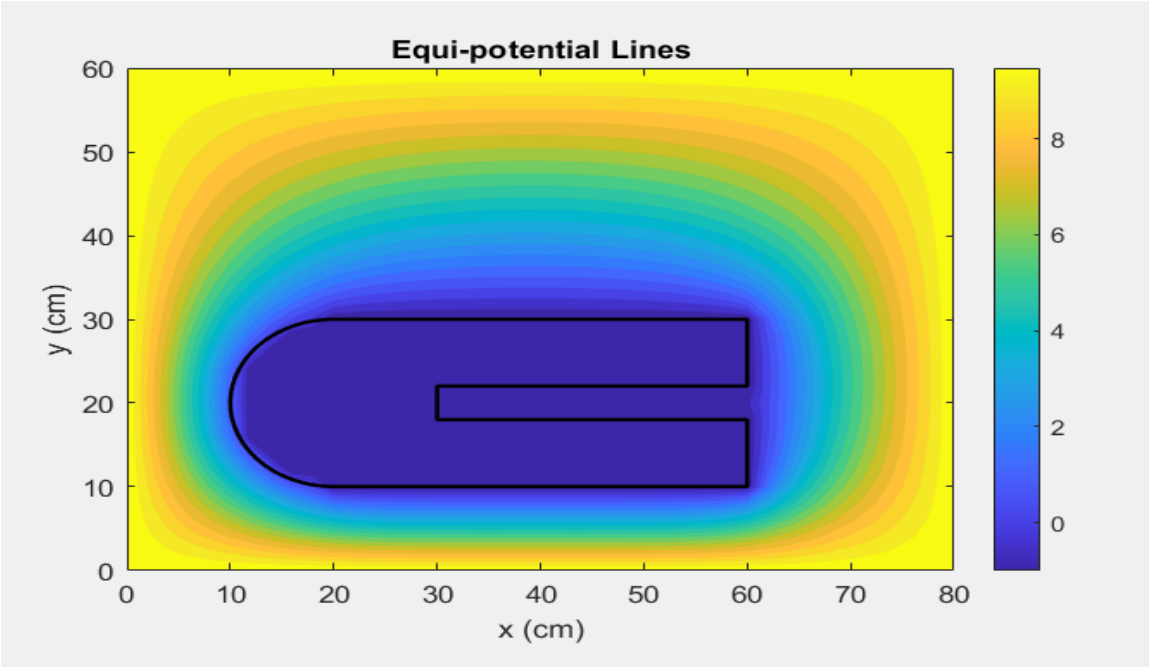


Figure 3 Equi-potential Lines with step size 2 cm

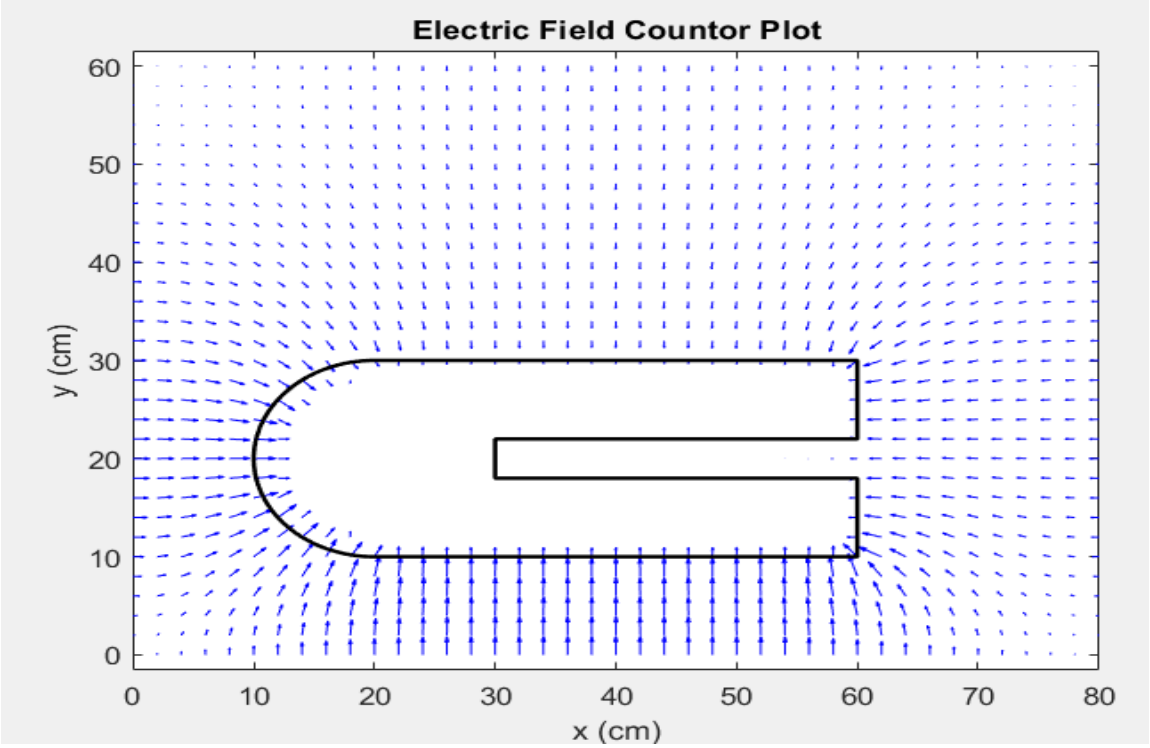


Figure 4 Electric Field Countor Plot size 2cm

3.5 QUESTION 4

Decrease the step size to 1 cm, 0.5 cm and 0.25 cm and repeat parts 1-3.

3.5.1 Updated matlab code just with grid sizes(1 cm):

```
dx = 1; % Grid step in cm
dy = 1;
```

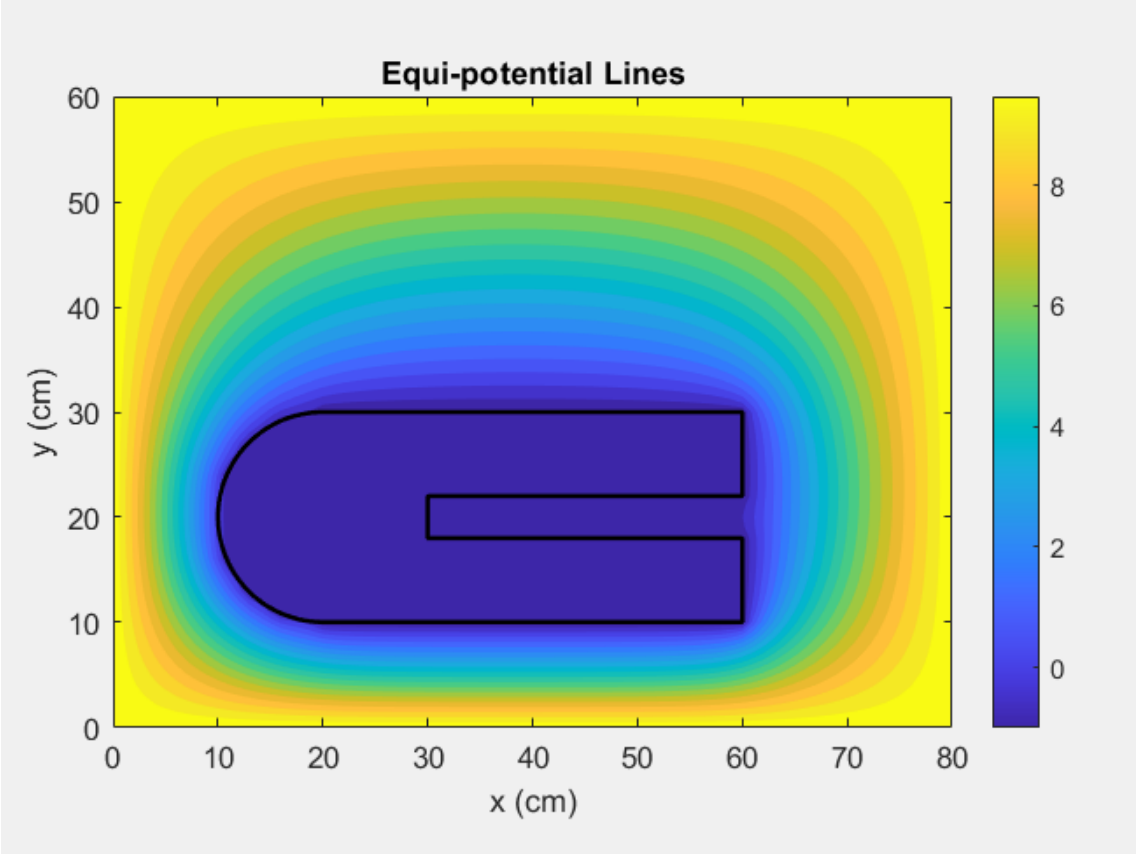


Figure 5 Equi-potential Lines with step size 1 cm

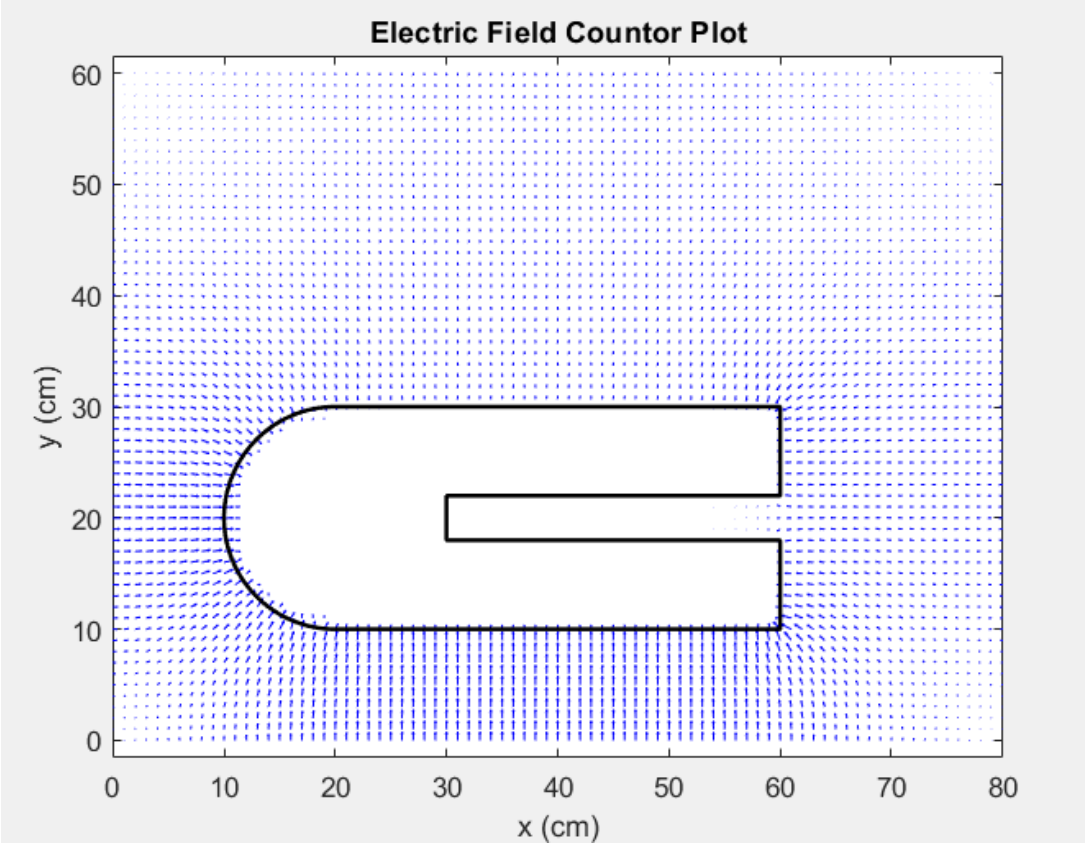


Figure 6 Electric Field Countor Plot size 1 cm

3.5.2 Updated matlab code just with grid sizes (0.5 cm):

```
dx = 0.5; % Grid step in cm
dy = 0.5;
```

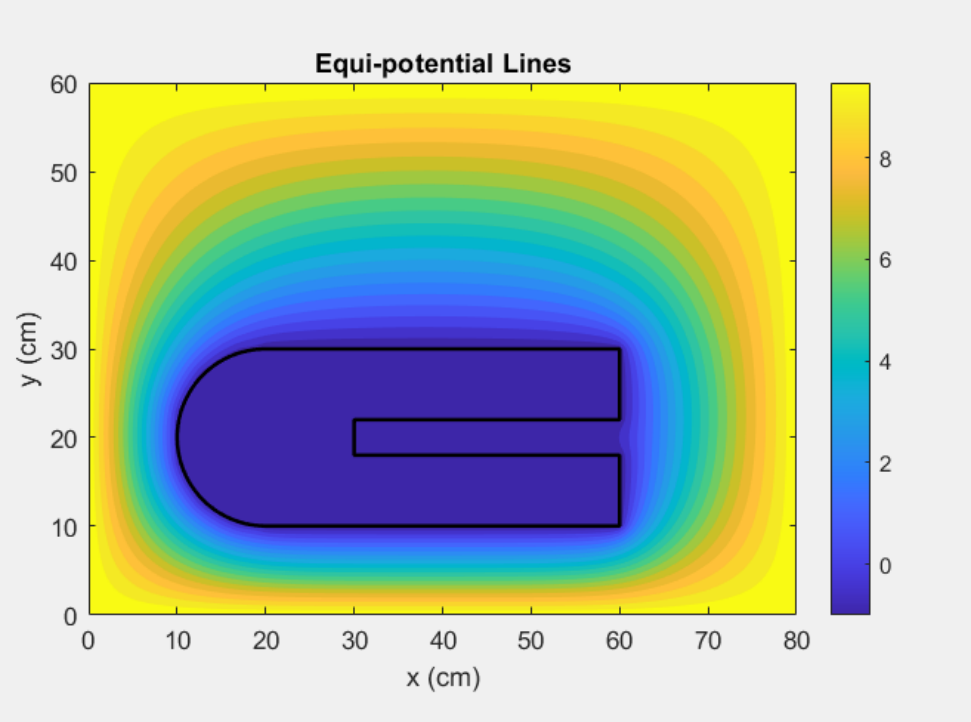


Figure 7 Electric Field Countor Plot size 1 cm

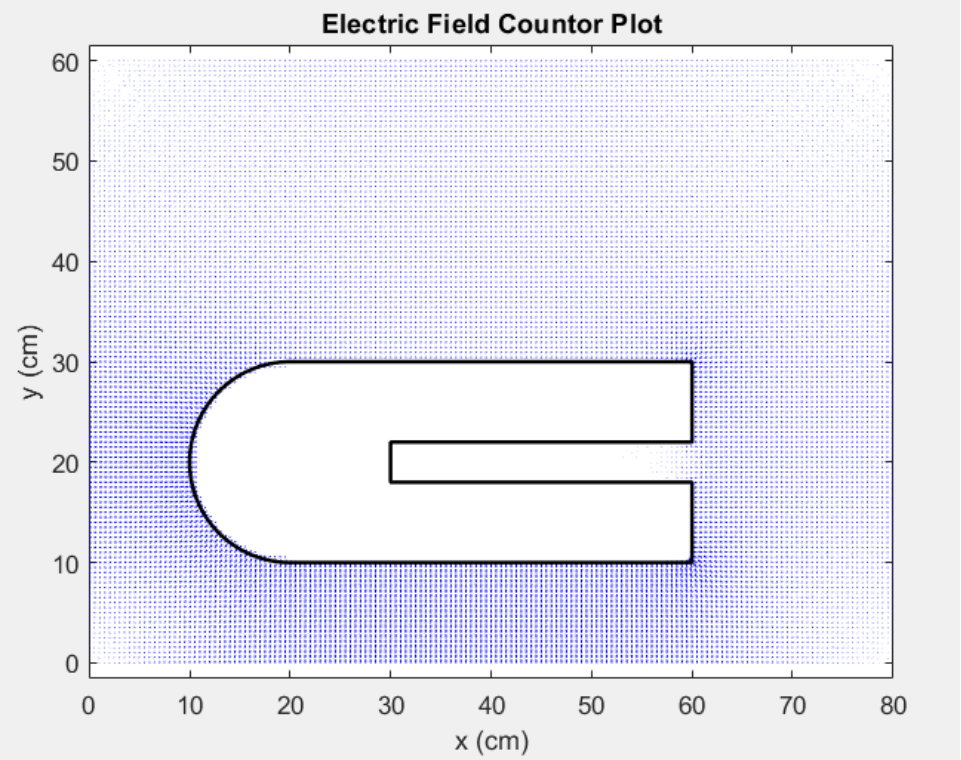


Figure 8 Electric Field Countor Plot size 0.5 cm

3.5.3 Updated matlab code just with grid sizes (0.25 cm):

```
dx = 0.25; % Grid step in cm
dy = 0.25;
```

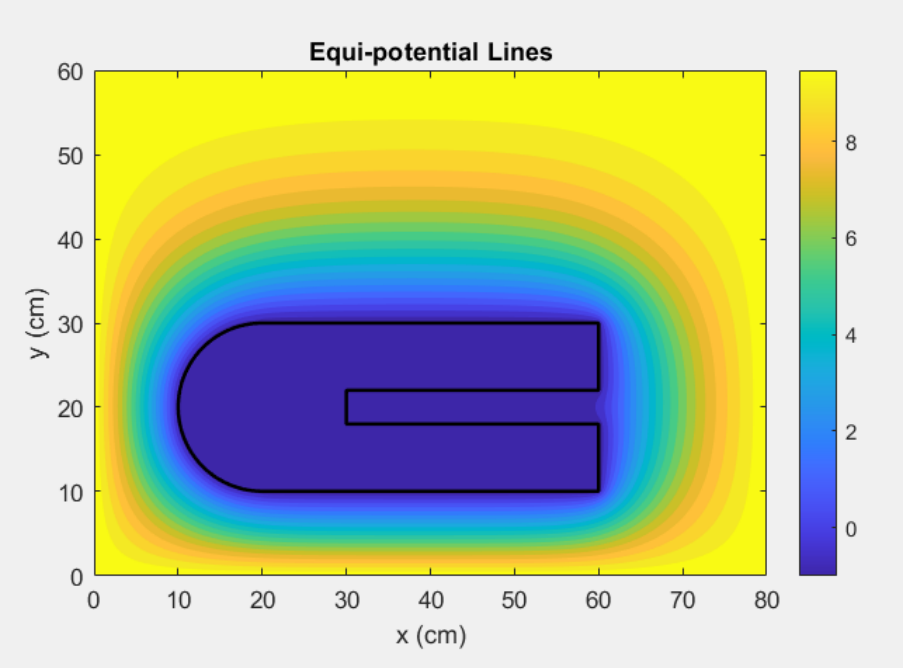


Figure 9 Equi-potential Lines with step size 0.25 cm

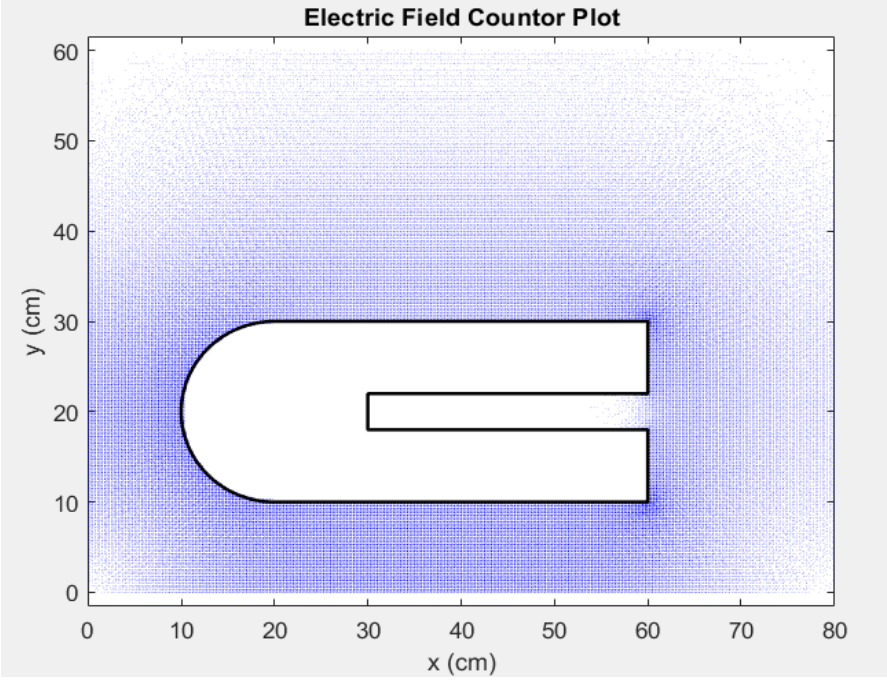


Figure 10 Electric Field Countor Plot size 0.25 cm

3.6 QUESTION 5

Perform error analysis by observing how the value of the potential at specific point(s) inside the structure change as the step size is reduced. Produce a plot and give comments.

If we think of the structure on the XY Plane. We can select one point for analysis. We select (*36*,*36*) point for observing on the change of voltage values.

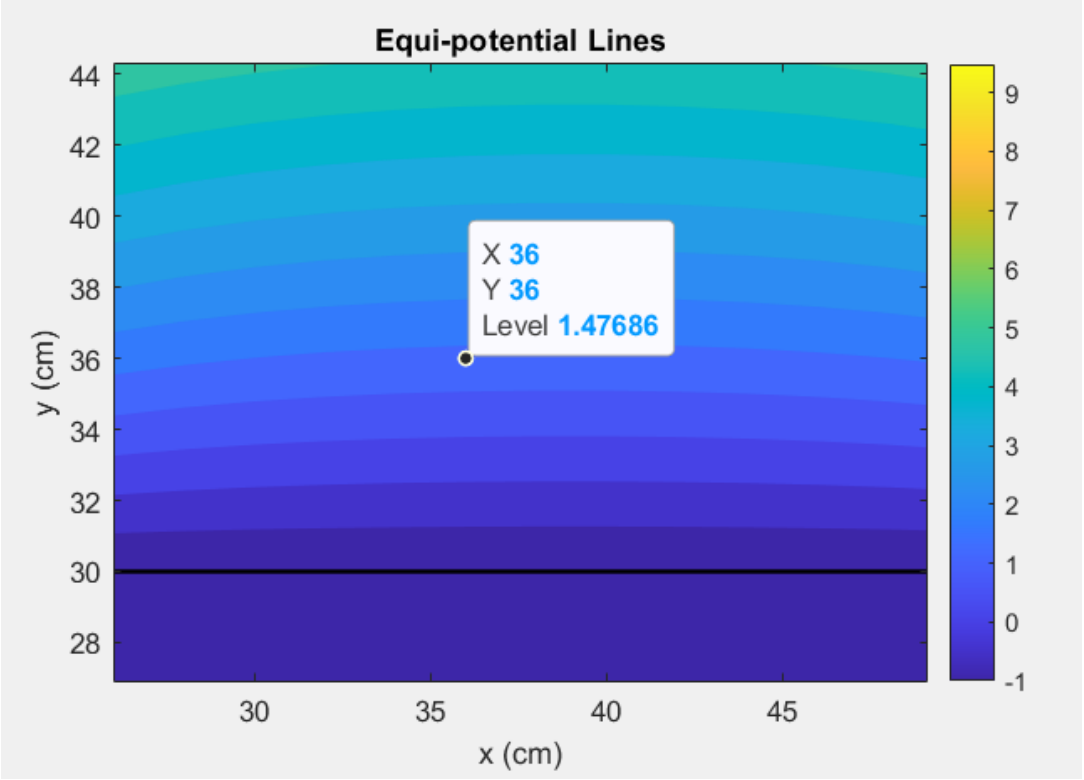


Figure 11 Error Analysis size 2 cm

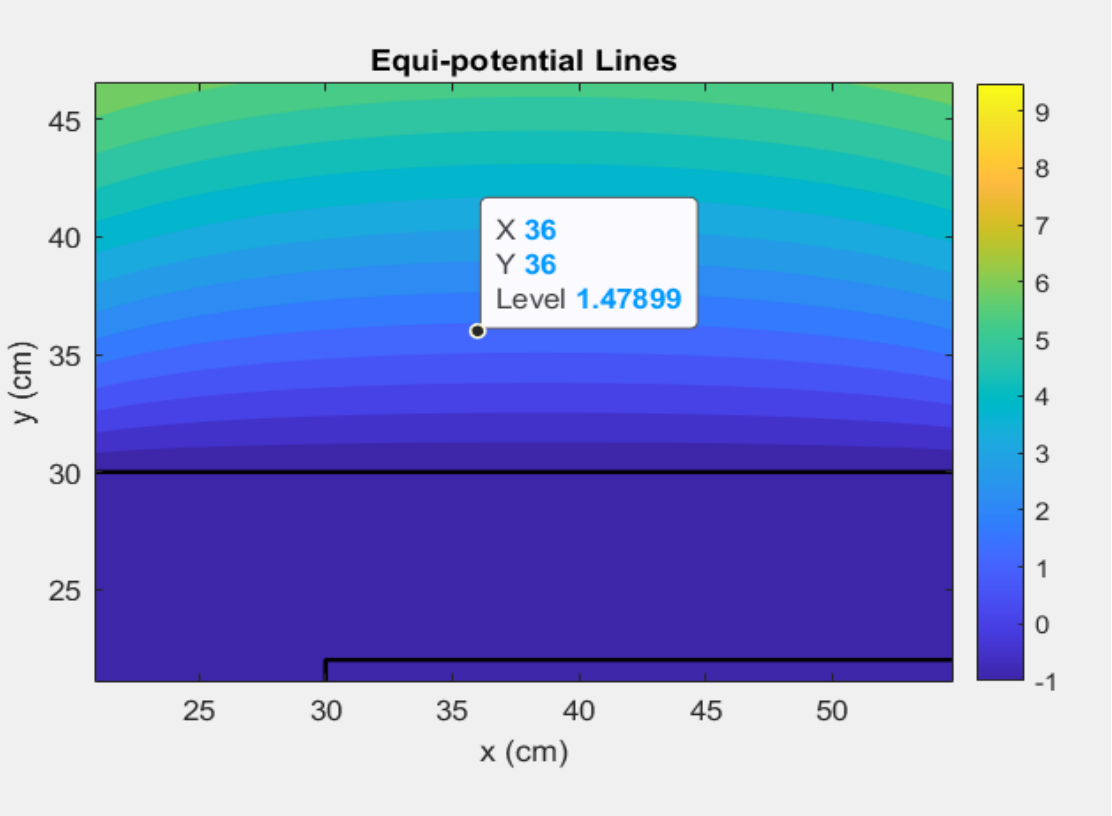


Figure 12 Error Analysis size 1 cm

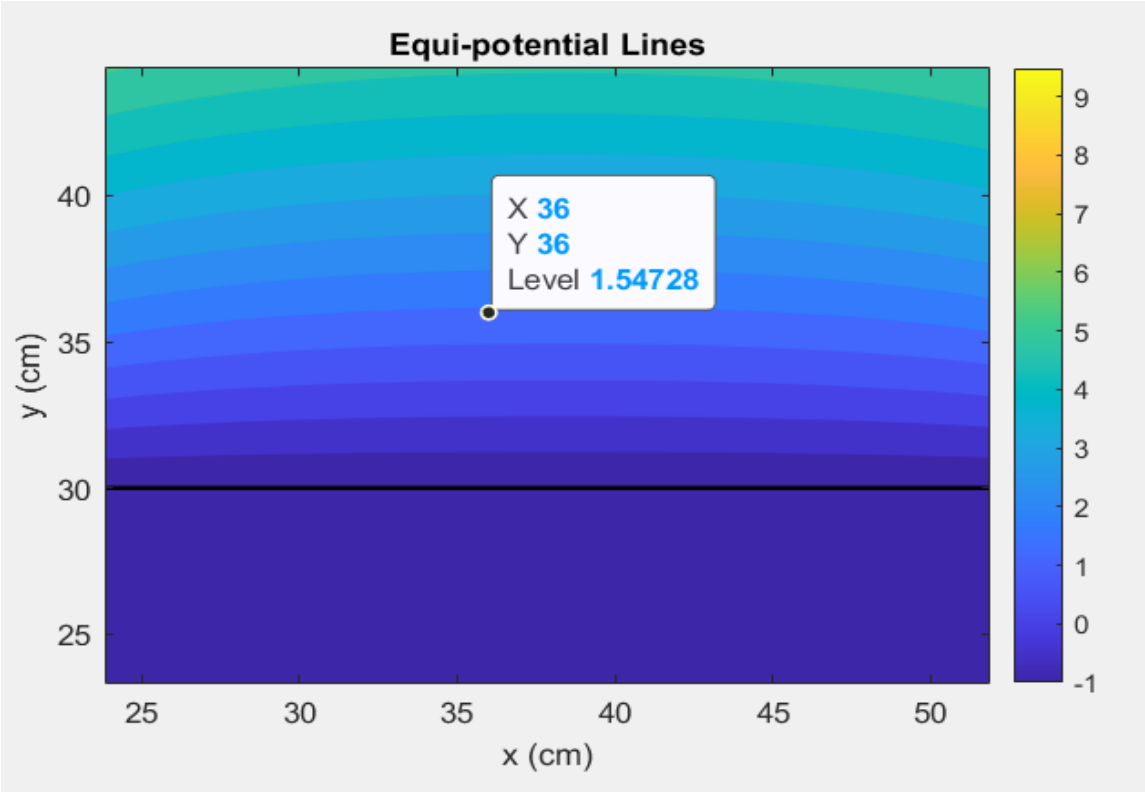


Figure 13 Error Analysis size 0.5 cm

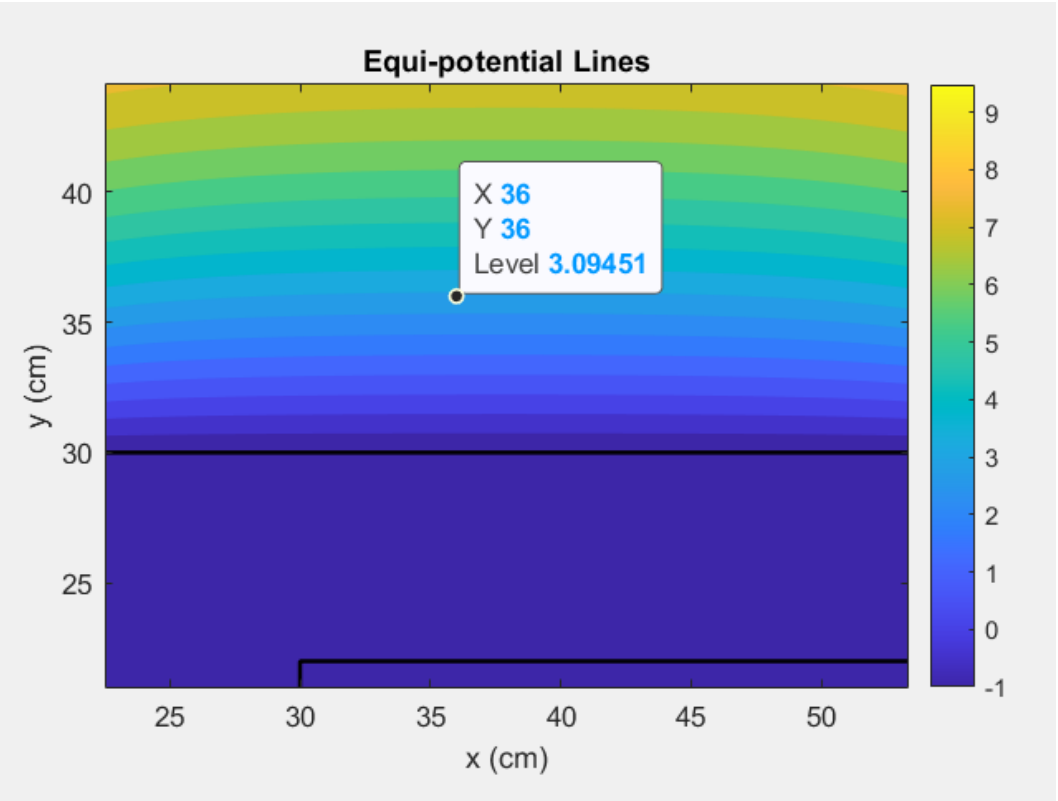


Figure 14 Error Analysis size 0.25 cm

Table 1 Error Analysis

	Step Size (cm)	Potential Value(V)	Difference from Reference(value in 0.25 step size)
1	0.25	3.09451	0.0
2	0.50	1.54728	1.54723
3	1	1.47899	1.61552
4	2	1.47686	161765

3.6.1 Matlab Coding of Error Analysis

```
clc; clear; close all;

% Step sizes and corresponding potential values
step_sizes = [0.25, 0.5, 1, 2]; % Increasing order of step sizes
potential_values = [3.09451, 1.54728, 1.47899, 1.47686];

% Reference potential (for step size 0.25)
reference_potential = 3.09451;

% Calculate differences with respect to the reference potential
potential_differences = abs(potential_values -
reference_potential);
```

```
% Plot the differences versus step sizes
figure;
plot(step_sizes, potential_differences, '-o', 'LineWidth', 1.5);
xlabel('Step Size (cm)');
ylabel('Difference in Potential (V)');
title('Error Analysis: Difference in Potential vs Step Size');
grid on;

% Add comments to the plot
for i = 1:length(step_sizes)
    text(step_sizes(i), potential_differences(i), sprintf('%.4f',
potential_differences(i)), ...
        'VerticalAlignment', 'bottom', 'HorizontalAlignment',
'right');
end

% Display results in the command window
disp('Step Size (cm) | Potential Difference (V)');
disp([step_sizes', potential_differences']);
```

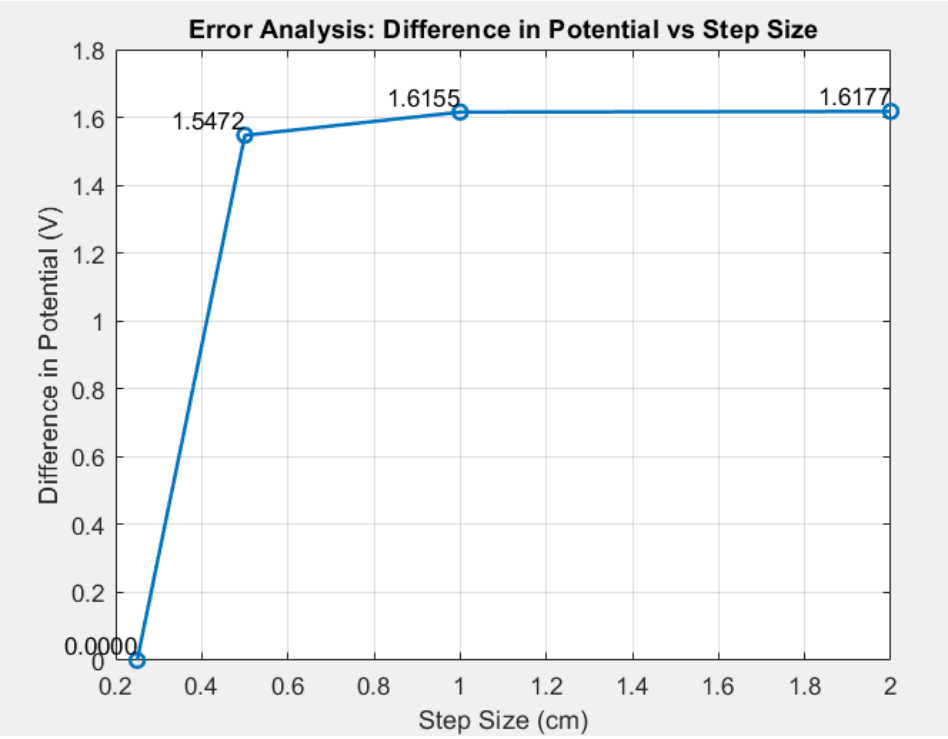


Figure 15 Error Analysis

3.6.2 Comment of Question 5

The error analysis shows that decreasing the step size improves accuracy, with the potential value converging to the reference (3.09451V at 0.25 cm). Smaller steps yield more accurate results, but beyond 1 cm, the error stabilizes (~1.617V), indicating limited improvement with coarse grids. This highlights the trade-off between accuracy and computational efficiency, where finer grids offer precision but demand more resources.

3.7 QUESTION 6

Make one modification to the given structure (unique to your team) by introducing some features inside the basic structure. Find the new potential and field distributions. Give comments.

3.7.1 Matlab Coding of Question 6

```
clc; clear; close all;

% Grid setup
dx = 2; % Grid step in cm
dy = 2;
x = 0:dx:80; % x-coordinates
y = 0:dy:60; % y-coordinates
[X, Y] = meshgrid(x, y);

% Initialize potential matrix
V = 5 * ones(length(y), length(x)); % Entire area with 5V
initially

% Define boundary conditions for the closed region (-3V)
theta = linspace(pi/2, 3*pi/2, 100); % Semi-circle (counter-
clockwise direction)
r = 10; % Radius of semi-circle
x_circle = 20 + r * cos(theta); % Semi-circle x-coordinates
y_circle = 20 + r * sin(theta);

r2 = 5.5;
x_circle2 = 30 + r2 * cos(theta);
y_circle2 = 18.5 + r2 * sin(theta);

r3 = 2;
x_circle_UP = 60 + r3 * cos(theta);
y_circle_UP = 26 + r3 * sin(theta);

r4 = 1;
x_circle_DOWN = 60 + r4 * cos(theta);
y_circle_DOWN = 12 + r4 * sin(theta);

% Update the closed region with new coordinates
x_closed = [x_circle, 20, 30, 30, 60, fliplr(x_circle_DOWN), 60,
30, fliplr(x_circle2), 30, 60, fliplr(x_circle_UP), 60, 30 ,30,
20];
y_closed = [y_circle, 10, 10, 11, 11, fliplr(y_circle_DOWN), 13,
13, fliplr(y_circle2), 24, 24, fliplr(y_circle_UP), 28, 28 ,30,
30];

% Identify the points inside the closed region
mask = inpolygon(X, Y, x_closed, y_closed);

% Apply boundary conditions
V(mask) = -3; % Inside the closed region, potential is -3V
V(~mask & (X == 0 | X == 80 | Y == 0 | Y == 60)) = 5; % Boundary
conditions on outer edges

% Solve Laplace's equation using iterative FD method
tolerance = 1e-4;
```

```
max_iter = 5000;
error = Inf;
iter = 0;

while error > tolerance && iter < max_iter
    V_old = V;

    % Update potential using finite difference (excluding boundary
conditions)
    for i = 2:length(y)-1
        for j = 2:length(x)-1
            if ~mask(i, j) % Update only for points outside the
fixed potential regions
                V(i, j) = 0.25 * (V_old(i+1, j) + V_old(i-1, j) +
V_old(i, j+1) + V_old(i, j-1));
            end
        end
    end

    % Compute error
    error = max(max(abs(V - V_old)));
    iter = iter + 1;
end

% Compute electric field (E = -grad(V))
[Ex, Ey] = gradient(-V, dx, dy);

% Plot the potential distribution with contour lines
figure;
contourf(X, Y, V, 20, 'LineColor', 'none');
colorbar;
title('Equi-potential Lines');
xlabel('x (cm)');
ylabel('y (cm)');
axis equal;
hold on;

% Overlay the main closed region boundary
plot(x_closed, y_closed, 'k-', 'LineWidth', 1.5);

% Plot the electric field as a vector plot
figure;
quiver(X, Y, Ex, Ey, 'b');
title('Electric Field');
xlabel('x (cm)');
ylabel('y (cm)');
axis equal;
hold on;

% Overlay the main closed region boundary
plot(x_closed, y_closed, 'k-', 'LineWidth', 1.5);
```

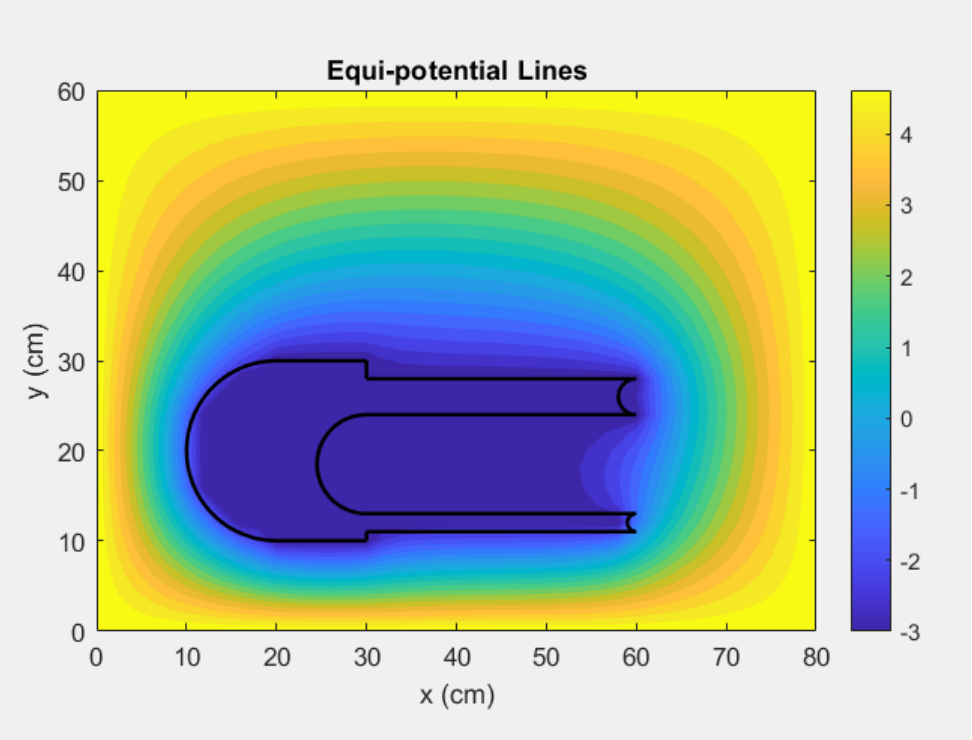


Figure 16 Equi-potential Lines with step size 2 cm

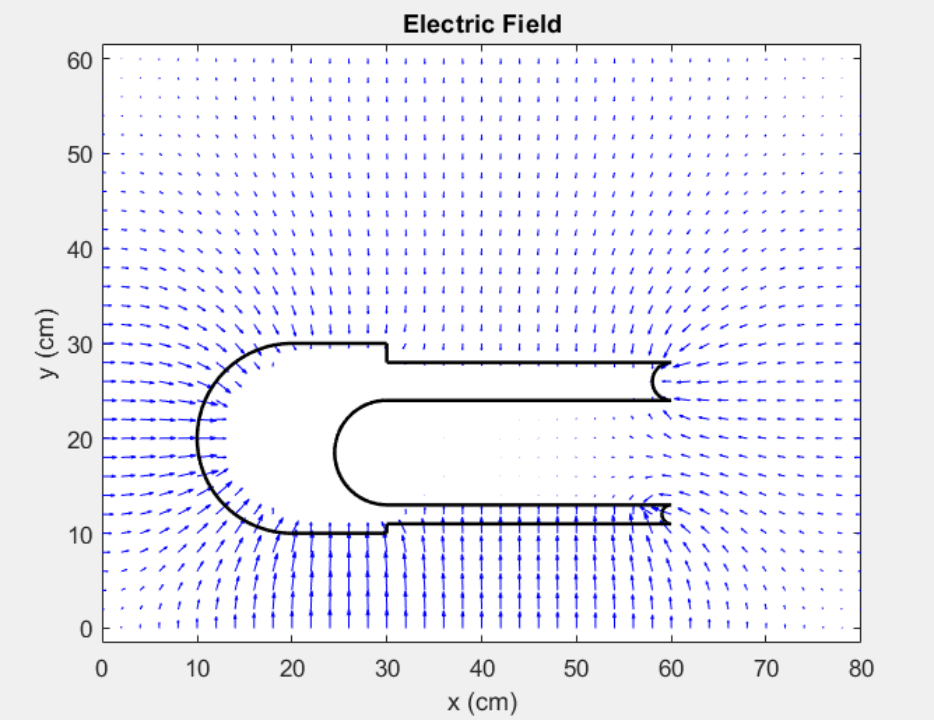


Figure 17 Electric Field Countor Plot size 2 cm

3.7.2 Updated matlab code just with grid sizes(1 cm):

```
dx = 1; % Grid step in cm
dy = 1;
```

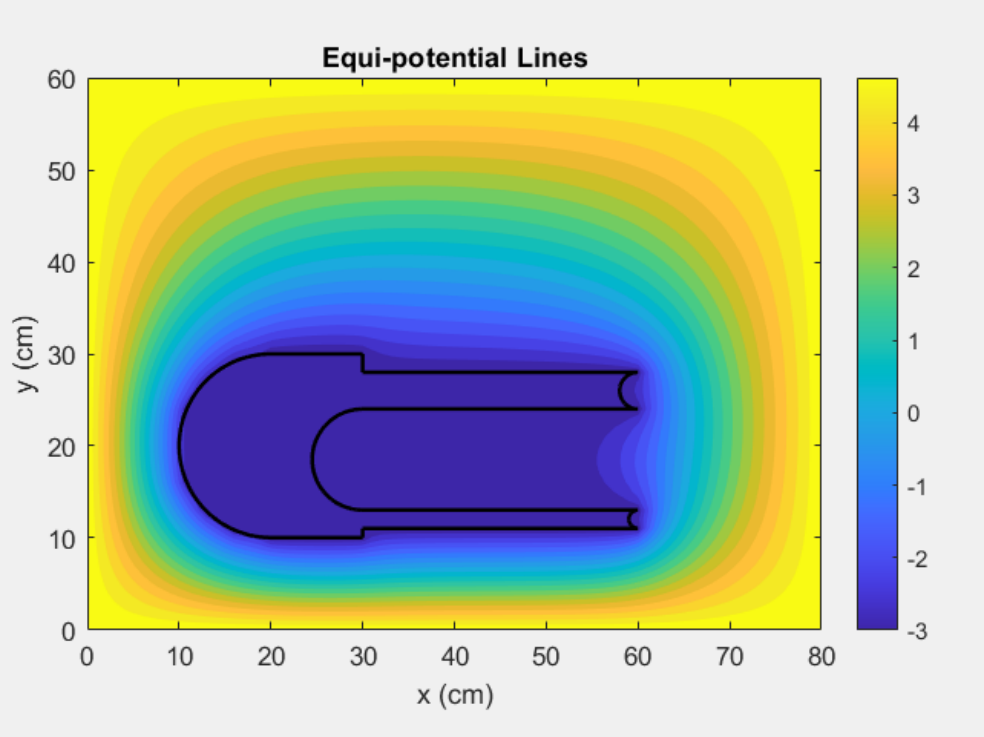


Figure 18 Equi-potential Lines with step size 1 cm

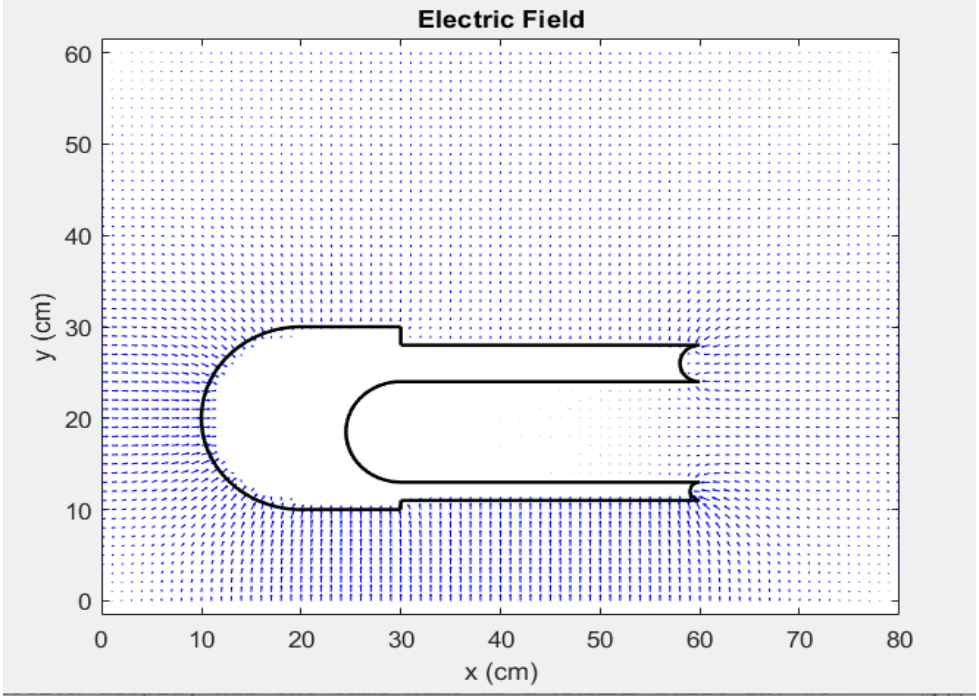


Figure 19 Electric Field Countor Plot size 1 cm

3.7.3 Updated matlab code just with grid sizes(0.5 cm):

```
dx = 0.5; % Grid step in cm
dy = 0.5;
```

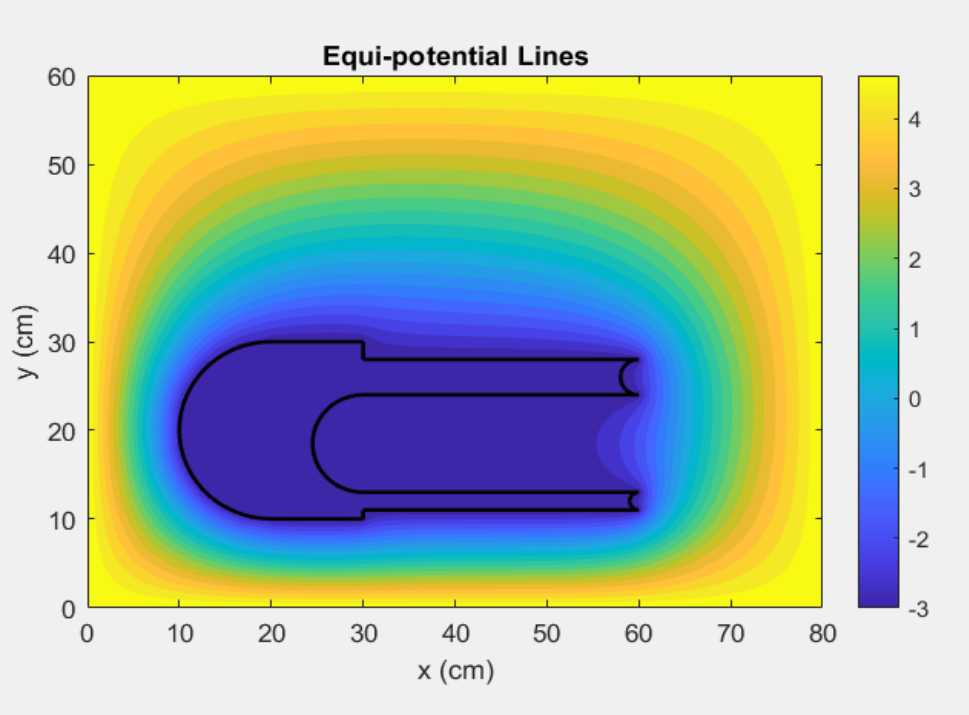


Figure 20 Equi-potential Lines with step size 0.5 cm

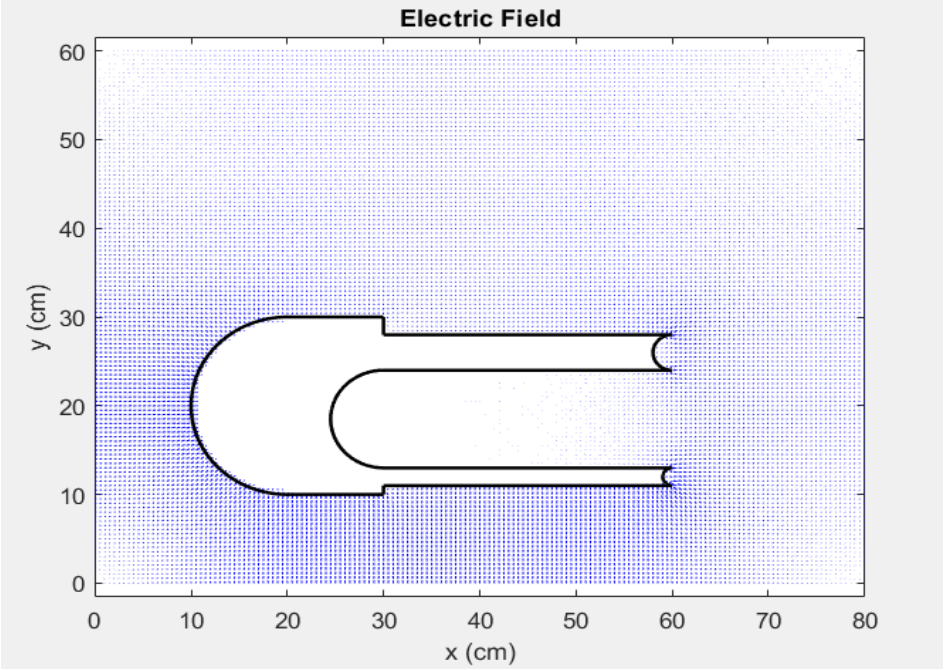


Figure 21 Electric Field Countor Plot size 0.5 c

3.7.4 Updated matlab code just with grid sizes(0.25 cm):

```
dx = 0.25; % Grid step in cm
dy = 0.25;
```

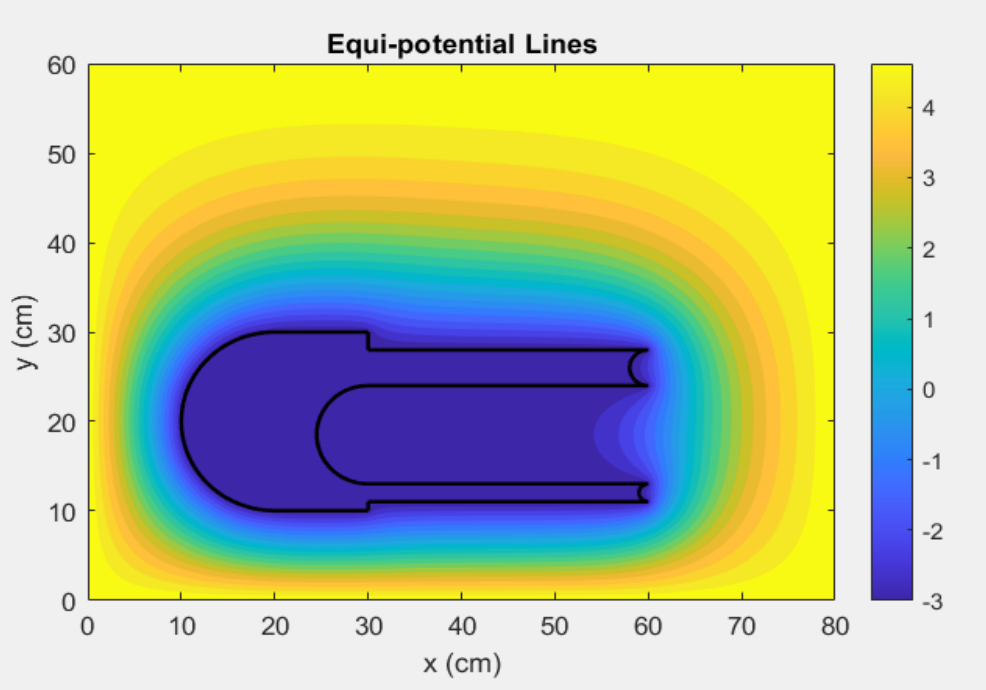


Figure 22 Equi-potential Lines with step size 0.25 cm

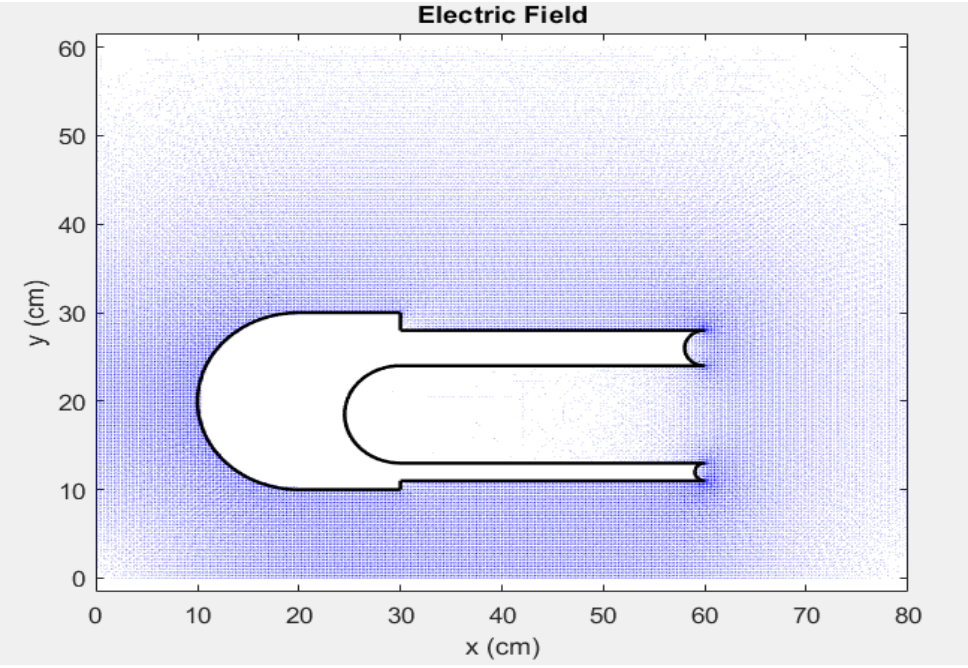


Figure 23 Electric Field Countor Plot size 0.25 cm

3.7.5 Comment of Question 6

The electric field and equipotential line distributions differ significantly between the given structure and the unique structure. In the given structure, the electric field is symmetric and exhibits relatively uniform gradients around the boundaries, with smooth, evenly distributed intensities across the region. Similarly, the equipotential lines are smooth, symmetric, and evenly spaced, reflecting a gradual and balanced potential change due to the simpler geometry. In contrast, the unique structure introduces significant asymmetry, with the electric field showing localized regions of high intensity near sharp edges and narrow extensions. The equipotential lines in the unique structure become denser and more irregular near these areas, indicating sharper potential gradients and a less uniform distribution. These differences highlight the impact of the unique structure's geometry, which creates sharper gradients and concentrated field and potential variations compared to the given structure.

4. DRIVE LINK OF CODES

You can find all necessary files the link below:

https://drive.google.com/drive/folders/1lIJCm74b4Z3of_O9mVuY_kU1xxKMSYN4?usp=sharing

5. REFERENCES

[1] Sadiku, Matthew N. O. *Elements of Electromagnetics*. 6th ed., Oxford University Press, 2015.

[2] MATLAB. *Documentation.* The MathWorks, Inc. Accessed December 2024. Available at: https://www.mathworks.com/help